A SUBSPACE OF K TO THE N MEETS THE BASE FIELD IF AND ONLY IF IT MEETS EVERY SUBFIELD OF A FILTERING FAMILY

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Allowed standard prerequisites (editorial).

- The family of subfields is filtering downwards and has intersection k, with k contained in every member. hypotheses (1), (2) and (3) of the statement, source lines 11419-11424. Hypothesis (3) produces the smaller subfield K_alpha' used in the last step, and (2) identifies the intersection of all subfields below K_alpha with k.
- Intersection with a subspace over a smaller field: if V cap K_alpha to the n is spanned by a single vector with first coordinate 1, then elements of V over a subfield are scalar multiples of it. used at proof lines 11443-11447. Shows that V cap K_alpha to the n equals K_alpha v, which forces the later comparison with K_alpha'.
- Induction hypothesis applied to subspaces of K to the n-1 consisting of vectors whose first coordinate is zero. used at proof lines 11437-11440 to choose alpha with V cap 0 plus K_alpha to the n-1 equal to zero. Produces the subfield that makes the first coordinate of the chosen vector nonzero.

Source reference. Original lemma 07P3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9783–9783. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let K/k be a field extension. Let $\{K_\alpha\}_{\alpha\in A}$ be a collection of subfields of K with the following properties*

(1) *$k \subset K_\alpha$ for all $\alpha \in A$,*
(2) *$k = \bigcap_{\alpha\in A} K_\alpha$,*
(3) *for $\alpha, \alpha' \in A$ there exists an $\alpha'' \in A$ such that $K_{\alpha''} \subset K_\alpha \cap K_{\alpha'}$.*

Then for $n \geq 1$ and $V \subset K^{\oplus n}$ a K-vector space we have $V \cap k^{\oplus n} \neq 0$ if and only if $V \cap K_\alpha^{\oplus n} \neq 0$ for all $\alpha \in A$.

Proof. By induction on n. The case $n = 1$ follows from the assumptions. Assume the result proven for subspaces of $K^{\oplus n-1}$. Assume that $V \subset K^{\oplus n}$ has nonzero intersection with $K_\alpha^{\oplus n}$ for all $\alpha \in A$. If $V \cap 0 \oplus k^{\oplus n-1}$ is nonzero then we win. Hence we may assume this is not the case. By induction hypothesis we can find an α such that $V \cap 0 \oplus K_\alpha^{\oplus n-1}$ is zero. Let $v = (x_1, \ldots, x_n) \in V \cap K_\alpha^{\oplus n}$ be a nonzero element. By our choice of α we see that x_1 is not zero. Replace v by $x_1^{-1}v$ so that $v = (1, x_2, \ldots, x_n)$. Note that if $v' = (x'_1, \ldots, x'_n) \in V \cap K_\alpha^{\oplus n}$, then $v' - x'_1 v = 0$ by our choice of α. Hence we see that $V \cap K_\alpha^{\oplus n} = K_\alpha v$. If we choose some α' such that $K_{\alpha'} \subset K_\alpha$, then we see that necessarily $v \in V \cap K_{\alpha'}^{\oplus n}$ (by the same arguments applied to α'). Hence

$$x_2, \ldots, x_n \in \bigcap\nolimits_{\alpha'\in A, K_{\alpha'}\subset K_\alpha} K_{\alpha'}$$

which equals k by (2) and (3). □